

Journal Trend Analyzer - Technical Report

Project: `journal_trend_analysis`

Framework: Flutter / Dart SDK `^3.12.0`

Firebase project: `journaltrend-8cc04`

Last updated: July 17, 2026

1. Overview

Journal Trend Analyzer is a Flutter application for researching scientific publications and journals through the OpenAlex REST API. It combines search, journal exploration, keyword analytics, trend visualisation, report export, and Firebase-backed account features.

The app supports two access levels:

- **Guest users** can browse Home, Journals, Keywords, papers, dashboards, heatmaps, and author networks.
- **Google-authenticated users** can also save papers, view saved papers, enable research-update notifications, export PDF reports, and use the authenticated Profile features.

The project follows a practical Clean Architecture style: providers compose dependencies, repositories hide data sources, use cases hold reusable calculations, and screens render Riverpod state.

2. Navigation

The router uses a `ShellRoute` with a persistent four-item `NavigationBar`.

```

ShellRoute
|-- /home           Home / trending publications
|-- /journals       Journal search and ranking
|-- /keywords       Keyword search with four analysis tabs
`-- /profile        Guest sign-in prompt or authenticated Profile

Standalone routes
|-- /login
|-- /publication/:id
|-- /journals/:id
`-- /profile/bookmarks

```

The `/keywords` screen contains the **Papers**, **Dashboard**, **Heatmap**, and **Network** analysis tabs. Publication and journal detail routes receive their loaded entity through `GoRouter` route extras.

3. Current Source Structure

```

journal_trend_analysis/
|-- lib/
|   |-- core/                API constants, router, theme,
format/export helpers
|   |-- data/                OpenAlex API client, remote/local
data sources, models, repositories
|   |-- domain/              entities, repository contracts,
analysis/search use cases
|   |-- firebase/            Auth, Analytics, Crashlytics, FCM,
Remote Config, PDF report services
|   |-- presentation/
|   |   |-- providers/        Riverpod dependency injection and UI
state
|   |   |-- screens/          Home, Journals, Keywords, Profile,
detail and saved-paper screens
|   |   `-- widgets/          shared charts, cards,
loading/error/empty states
|   |-- firebase_options.dart
|   |-- main.dart
|-- android/                  Firebase plugins, FCM channel, Patrol
instrumentation setup
|-- ios/                       Runner configuration and
GoogleService-Info.plist
|-- patrol_test/               Android end-to-end Patrol scenarios
|-- test/                      service and widget tests
|-- storage.rules               per-user PDF report protection
`-- pubspec.yaml

```

Important source files include:

- `lib/main.dart`: Firebase initialisation, Crashlytics error capture, background FCM handler registration, and Remote Config subscription.
 - `lib/core/router/app_router.dart`: routing and guest protection for saved papers.
 - `lib/presentation/screens/keywords_screen.dart`: topic search input and the integrated analysis tabs.
 - `lib/presentation/screens/profile_screen.dart`: authenticated features, notification history, PDF export, and compact Crashlytics test controls.
 - `lib/firebase/notification_service.dart`: FCM permission/topic subscription and local message history.
-

4. OpenAlex Research Features

Home

The Home tab presents trending publications, domain categories, and yearly publication counts. It uses OpenAlex work data and local derived state.

Journals

The Journals tab supports journal discovery and sorting by papers, citations, author-related statistics, or recent activity. Selecting a journal opens its details and publications.

Keywords and Analysis

Users can enter free-text queries or select an OpenAlex hierarchy item (domain, field, subfield, or topic). A selected search drives four connected views:

- **Papers**: paginated publication results, local sorting, and detail navigation.
- **Dashboard**: publication KPIs, yearly trends, top journals, and top authors.
- **Heatmap**: country or institution research distribution.
- **Network**: collaboration graph derived from the loaded publications.

Publication detail shows title, authors, journal, year, citations, topics, abstract, DOI/external links, and a bookmark action for signed-in users.

OpenAlex endpoints

Use case	Endpoint	Main parameters
Free-text publication search	GET /works	search, page, per_page, sort=relevance_score:desc
Topic-filter search	GET /works	filter, page, per_page, citation sorting
Trending publications	GET /works	recent date/type filter, page, citation sorting
Journal browse/search	OpenAlex sources endpoint	query, page, sort, journal filter
Topic hierarchy/autocomplete	domains, fields, subfields, topics endpoints	parent filter or search text
Heatmap and yearly aggregates	GET /works	group_by or yearly grouping

`OpenAlexApiClient` owns the Dio configuration, including timeouts and the polite-pool user agent. Models are converted to framework-independent domain entities by repository implementations.

5. State and Architecture

```
Screen interaction
-> Riverpod state provider
-> Future/Stream provider
-> use case or repository
-> OpenAlex or Firebase service
-> domain entity / UI state
-> reactive screen rebuild
```

Key state includes `searchQueryProvider`, `selectedTopicFilterProvider`, pagination, sort choices, journal state, bookmarks, authentication state, and Remote Config

limits. Derived providers calculate dashboard summaries, top authors, top journals, trends, and network/heatmap input from the current publication set.

The project uses:

- **Repository pattern:** OpenAlex publication and journal repository contracts hide data-source details.
 - **Dependency injection:** Riverpod providers construct API clients, data sources, repositories, and services.
 - **Reactive state:** asynchronous values expose loading, error, empty, and data states to screens.
 - **Local analysis:** dashboard, ranking, heatmap inputs, and network relationships are calculated from loaded data where appropriate.
-

6. Firebase Features

Google Authentication and Guest Access

`AuthService` authenticates with Google Sign-In and exchanges the Google ID token for Firebase Auth credentials. Authentication state controls Profile access and protects the saved-paper route. Sign-out clears both Firebase Auth and the Google session.

Guests remain able to research content. The Profile tab explains that saving, notifications, and PDF reports require a sign-in.

Analytics and Crashlytics

Firebase Analytics records key research and account events through `AnalyticsService`. Crashlytics collection is enabled at startup; Flutter framework errors and uncaught platform errors are recorded.

The Profile screen exposes only small icon controls for testing:

- record a handled demonstration exception;
- confirm and trigger a fatal test crash.

These controls are for development verification and are not presented as primary product features.

Remote Config

Remote Config is deliberately hidden from end users. `RemoteConfigService` supplies these parameters:

Parameter	Default	Runtime effect
<code>max_journals_displayed</code>	10	page size for journal lists
<code>max_keywords_displayed</code>	10	keyword autocomplete limit and publication result page size

Values are clamped to 1-100. The service fetches and activates at launch (with a one-hour minimum fetch interval), retains defaults if offline, and listens for Firebase real-time config updates. Screens/providers consume the resulting limits, so configuration changes affect actual API result sizes rather than merely showing diagnostic values in Profile.

Cloud Messaging and Notification Center

Authenticated users can enable notifications from Profile. The app requests platform permission, checks the APNs token on iOS, and subscribes the device to the FCM topic:

```
journaltrend_updates
```

Firebase Console campaigns can target this topic, so testing does not require copying an individual FCM token. Incoming messages are stored locally in `SharedPreferences`, deduplicated by message ID, and limited to the newest 50 entries. Tapping **Notification Center** opens a bottom sheet with the stored message history and timestamps.

Android declares `POST_NOTIFICATIONS` and creates the high-importance `research_updates` channel used by FCM background notification messages. A foreground message is still handled by the app and stored in Notification Center; a separate local-notification implementation would be needed for a system heads-up banner while the app is open.

PDF Export and Firebase Storage

Authenticated users can export analytics for the active search topic. `ReportService` builds a PDF containing:

- dashboard summary and publication trend table;
- top journals and authors;
- most influential publications by citation count.

The PDF is stored at `reports/{uid}/report-{timestamp}.pdf`. The app retrieves its download URL and exposes an **Open PDF** action through `url_launcher`.

Storage rules allow read/write only when the Firebase Auth user ID matches the `{uid}` path segment:

```
reports/{userId}/{fileName}: request.auth.uid == userId
```

7. Android and iOS Notes

Android

- Package ID: `com.blackspace.journaltrend`.
- Firebase is configured through `android/app/google-services.json` and the Google Services/Crashlytics Gradle plugins.
- FCM uses the `research_updates` notification channel for new background notification campaigns.
- Patrol uses `PatrolJUnitRunner` with AndroidX Test Orchestrator and clears app data between test cases.

On Xiaomi/MIUI devices, Android instrumentation can be blocked or killed by device background restrictions. A standard Android emulator or a device without aggressive battery management is more reliable for unattended Patrol runs.

iOS

The repository contains `GoogleService-Info.plist` and the Google Sign-In URL scheme for bundle ID `com.blackspace.journaltrend`. Building and signing an IPA, configuring APNs, adding Push Notifications and Background Modes, and uploading an APNs key to Firebase require a macOS/Xcode environment and Apple Developer credentials. See `documents/IOS_FIREBASE_HANDOFF.md` for the remaining handoff steps.

8. Testing

Unit and widget tests

The `test/` directory covers:

- notification JSON persistence and refresh from local storage;
- Remote Config bounds/default behavior;
- PDF report service behavior;
- core widget/provider setup.

Patrol end-to-end tests

`patrol_test/lab03_e2e_test.dart` defines 11 scenarios:

1. Guest research access and Google Sign-In unlock.
2. Topic search returns publication results.
3. Publication details open.
4. Journals tab loads statistics.
5. Journal detail opens.
6. Keywords navigation exposes analysis tabs.
7. Dashboard displays trend, top journals, and top authors.
8. Authenticated Profile displays account features.
9. PDF report uploads and can be opened.
10. Remote Config diagnostics are not visible to users.
11. Logout returns to the guest profile.

Class com.blackspace.journaltrend.MainActivityTest

all > com.blackspace.journaltrend > MainActivityTest

11
tests

0
failures

0
skipped

4m58.51s
duration

100%
successful

Tests

Test	22031116BG - 13
runDartTest[lab03_e2e_test TC01 guest can browse and Google Sign-In unlocks profile]	passed (27.695s)
runDartTest[lab03_e2e_test TC02 topic search displays publication results]	passed (25.212s)
runDartTest[lab03_e2e_test TC03 publication opens complete details]	passed (26.658s)
runDartTest[lab03_e2e_test TC04 Journals tab displays journal statistics]	passed (12.415s)
runDartTest[lab03_e2e_test TC05 journal opens journal details]	passed (23.073s)
runDartTest[lab03_e2e_test TC06 Keywords tab displays keyword analysis navigation]	passed (12.323s)
runDartTest[lab03_e2e_test TC07 keyword dashboard displays required analysis]	passed (26.041s)
runDartTest[lab03_e2e_test TC08 Profile displays authenticated user information]	passed (28.269s)
runDartTest[lab03_e2e_test TC09 PDF report uploads and can be opened]	passed (58.248s)
runDartTest[lab03_e2e_test TC10 Remote Config diagnostics stay hidden from users]	passed (30.507s)
runDartTest[lab03_e2e_test TC11 logout returns to guest profile]	passed (28.072s)

Search/dashboard cases intentionally run as guests because research is a guest feature. Tests that require Profile actions perform Google Sign-In.

9. Main Libraries

Library	Role
flutter_riverpod	dependency injection and reactive state
go_router	shell and detail navigation
dio	OpenAlex HTTP client
fl_chart	trend charts
countries_world_map	geographic visualisation
firebase_core, firebase_auth, google_sign_in	Firebase initialisation and Google authentication
firebase_analytics, firebase_crashlytics	product analytics and crash reporting
firebase_remote_config	remotely managed result limits
firebase_messaging	FCM topic notifications
firebase_storage	user-scoped PDF uploads

Library	Role
pdf, path_provider, share_plus, url_launcher	report generation, file/link sharing, and external opening
shared_preferences	bookmarks and local notification history
patrol	Android end-to-end testing

10. Operational Notes

- New FCM campaigns only reach devices already subscribed to `journaltrend_updates`; completed campaigns are not replayed for later subscribers.
- Topic messaging broadcasts to all subscribed devices. Per-user targeting requires a trusted backend that stores/uses device tokens; clients should not subscribe themselves to sensitive user-ID topics.
- Remote Config defaults keep research usable when Firebase cannot be reached.
- PDF URLs are user-scoped by Firebase Storage rules, and the app opens the generated download URL only after a successful upload.
- The current iOS push/signing work remains a macOS/Xcode handoff item.

11. Screens

9:19



Home

- All Fields
- Physical Sciences
- Social Sciences



Trending

Showing 50 of 28.6M

- 1

Inline Hardware KV-Cache Compression for Long-Context Transformer Inference: An Architectural Case for a Memory-Path Comp...

2023 77.0K citations

Jakubův, Jan, Chvalovský, Karel, Goertzel, Zarathustra et...
- 2

Learning Multiple Layers of Features from Tiny Images

2024 25.5K citations

Alex Krizhevsky
- 3

Two new Later Stone Age sites from the Final Pleistocene in the Falémé Valley, eastern Senegal

2024 24.0K citations

- Home
- Journals
- Keywords
- Profile

9:19



Journals

 Search journals by name...

 Sort: Most Papers



228.2K journals found

- 1

Bulletin of Miscellaneous Information (Royal Gardens Ke...

 5.6M papers  518.2K citations  16.7M authors

5.6M papers >
- 2

The Journal of the Abraham Lincoln Association

 Abraham Lincoln Association

 3.5M papers  85.7K citations  10.6M authors

3.5M papers >
- 3

Medical Entomology and Zoology

 Japan Society of Medical Entomolo...

 1.9M papers  13.2M citations  5.6M authors

1.9M papers >
- 4

ChemInform

 Wiley

 487.7K papers  206.3K citations  1.5M authors

487.7K papers >
- 5

The Lancet

 Elsevier BV

 475.7K papers  10.4M citations  1.4M authors

475.7K papers >



Home



Journals



Keywords



Profile

9:19



Bulletin of Miscellaneous Information (R...



Bulletin of Miscellaneous Information (Royal Gardens Kew)

journal

GB

5.6M works 518.2K citations h-index: 241

2yr cite: 0.1 Since 0 ISSN: 0366-4457

Top Topics

Magnetic confinement fusion research

Particle accelerators and beam dynamics

Fusion materials and technologies

Filter by Topic

Authors

Papers

Sort: Papers

- 1** Sigmund Freud
3 papers • 4.1K citations >
- 2** David W. Orr
1 papers • 1.6K citations >
- 3** Jane Jacobs
1 papers • 3.3K citations >
- 4** John Neter
1 papers • 2.9K citations >
- 5** William Wasserman
1 papers • 2.9K citations >
- 6** Michael Kutner
1 papers • 2.9K citations >

9:19



Bulletin of Miscellaneous Information (R...



Bulletin of Miscellaneous Information (Royal Gardens Kew)

journal GB

5.6M works 518.2K citations h-index: 241

2yr cite: 0.1 Since 0 ISSN: 0366-4457

Top Topics

Magnetic confinement fusion research

Particle accelerators and beam dynamics

Fusion materials and technologies

Filter by Topic

Authors

Papers

Sort: Citations

- 1** ECDEU Assessment Manual for Psychopharmacology >
1976 8.8K citations
William Guy
- 2** Vibration problems in engineering >
1928 3.5K citations
S. Timoshenko
- 3** The Economy of Cities >
1969 3.3K citations
Jane Jacobs
- 4** Applied linear statistical models : regression, analysis of variance, and experimental designs >
1974 2.9K citations

9:20



Bulletin of Miscellaneous Information (R...



Bulletin of Miscellaneous Information (Royal Gardens Kew)

journal GB

5.6M works 518.2K citations h-index: 241

2yr cite: 0.1 Since 0 ISSN: 0366-4457

Top Topics

Magnetic confinement fusion research

Particle accelerators and beam dynamics

Fusion materials and technologies

Filter by Topic

Enter a Domain, Field, Subfield, or Topic keyword:



e.g. Machine Learning, Biolog...

Cancel

Apply

1

2

David W. Orr

1 papers • 1.6K citations

3

Jane Jacobs

1 papers • 3.3K citations

4

John Neter

1 papers • 2.9K citations

5

William Wasserman

1 papers • 2.9K citations

6

Michael Kutner

1 papers • 2.9K citations

9:20



Keywords

Artificial Intelligence

Artificial Intelligen...

Sort: Relevance

Artificial Intelligence (Subfield)

Papers

Dashboard

Heatmap

Network

Showing 7 of 4.2M results

1

Random Forests

Citations 127.6K

Year 2001

Machine Learning

Leo Breiman

Random forest

Mathematics

AdaBoost

Statistics

Tree (set theory)

Generalization

Support vector machine

Generalization error

Measure (data warehouse)

Artificial intelligence

Pattern recognition (psychology)

Computer science

Artificial neural network

Data mining

Combinatorics

Mathematical analysis

2

Long Short-Term Memory

Citations 99.1K

Year 1997

Neural Computation



Home



Journals



Keywords



Profile

Keywords

🔍

Artificial Intelligence

✕

📄 Artificial Intelligen... ▾

⚙️ Sort: Relevance ▾

🏷️ Artificial Intelligence (Subfield) ✕

Papers

Dashboard

Heatmap

Network

📅

Filter by year range

📄

4.2M

Total Papers

”

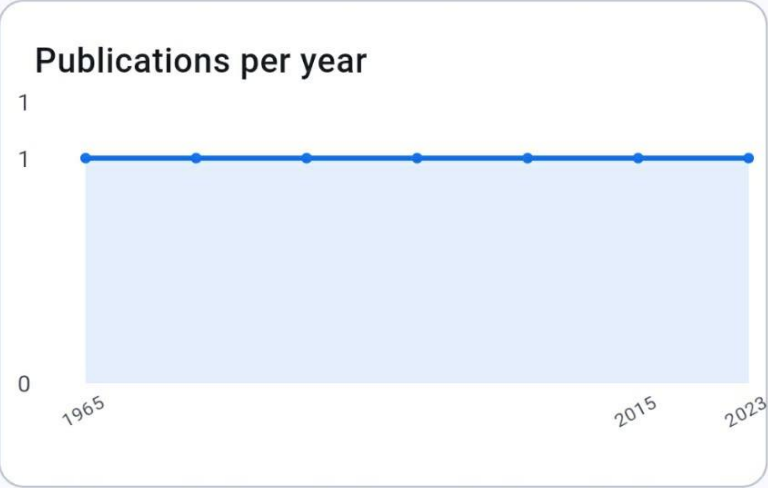
86464.0

Avg. Citations

📅

2023

Most Active Year



Rank by:

✓ Papers

Citations

Top Journals

9:20



Keywords

Artificial Intelligence

Artificial Intelligen...

Sort: Relevance

Artificial Intelligence (Subfield)

Papers

Dashboard

Heatmap

Network

✓ Countries

Institutions

200 countries contributing

Ranked by number of publications



Low



High

Tap a country to see details



Home



Journals



Keywords



Profile

9:20



Keywords

Artificial Intelligence

Artificial Intelligen...

Sort: Relevance

Artificial Intelligence (Subfield)

Papers

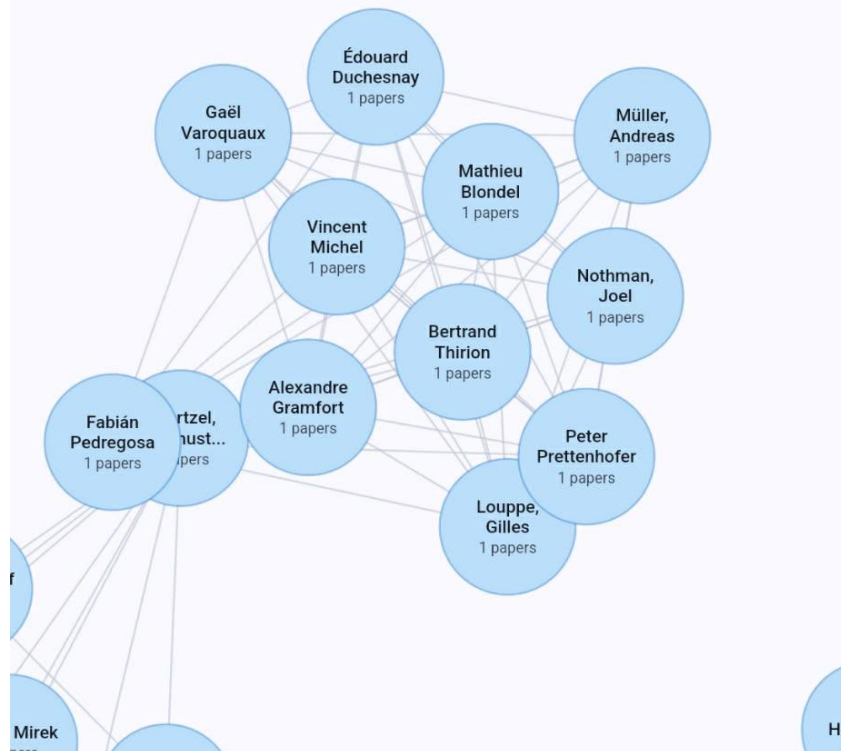
Dashboard

Heatmap

Network

Scale by Papers

Scale by Citations



Home



Journals



Keywords



Profile

9:21



Profile



Hoàng Long

kinglong112005@gmail.com

[Sign Out](#)



Saved Papers



Review and filter the publications you bookmarked.



Notification Center

Receive research update notifications.

[Enable](#)



Export PDF Report

Generate and upload analytics for the active topic.

[Export](#)



Home



Journals



Keywords



Profile